

BUSINESS APPLICATIONS OF AI

Preface — Why This Book Exists

Palm Beach Atlantic University · Graduate Program

Preface

Why This Book Exists — And Why It Is Different

There are already dozens of books about AI and business. Most of them are wrong in the same way.

They treat AI as a tool. A feature. Something you learn to use, add to your workflow, and then return to doing business as usual — only faster. They review the platforms, catalogue the use cases, and offer prompting tips as though the problem were a productivity deficit and the solution were a new application in your dock.

That is not this book.

This book begins from a different premise: AI is not a tool to learn — it is an operating environment to inhabit. The organizations winning right now are not the ones who added AI to their existing processes. They are the ones who rebuilt their processes inside AI, the way a fish does not add water to its habitat but lives inside it. Students who approach this course looking for a new skill to append to their résumé will be outcompeted by students who understand that the entire competitive landscape has restructured around a new medium — and that medium is AI.

That claim is not hype. It is the thesis of this course, and every chapter that follows will build the evidence for it and the capability to act on it.

What This Book Argues

The AI era is a structural inflection, not a cyclical one. Prior technologies — the internet, mobile, cloud — changed where business happened and how fast it moved. AI changes what can be done by whom at what cost. Interpretation, synthesis, persuasion, code generation, lead qualification, customer onboarding, competitive analysis — these were human cognitive tasks. They no longer are, exclusively.

This creates a barbell economy: extraordinary leverage for those who understand how to design systems that deploy AI well, and rapid displacement for those who do not. The asymmetry is not between large companies and small ones. It is between people who think architecturally about AI and people who think transactionally about it.

This book is designed to produce the former.

The Structure of This Course

This book spans eleven modules. They are not independent. Each one builds on the last, and by the final chapter, a student who has done the work — actually done the work, not read about it — will have produced something real:

- A live minimum viable product, built without traditional code, deployed to the internet
- A synthetic sales team: AI agents that prospect, qualify, and follow up on leads
- A marketing system: automated workflows that capture and convert customers
- An ethics framework for governing AI decisions with documented principles
- An agentic business architecture: a multi-agent system that executes business processes without requiring human intervention at every step

Not a slide deck. Not a reflection paper. A working system.

The sequence matters. Chapters 1 through 5 build strategic and analytical foundations: understanding AI disruption, validating markets, stress-testing unit economics, and pitching with precision. Chapters 6 through 8 shift to execution: vibe coding, synthetic employees, and martech systems. Chapters 9 through 11 operate at the highest level of complexity: the economics of information, AI ethics and governance, and the design of fully agentic enterprises. By the time you reach the final lab, you will have built something that Chapter 1 could only describe.

The Faith Dimension

This book was written at a Christian university, and it takes seriously what that means.

Technology is not neutral. Every system embeds assumptions about what matters, who counts, what efficiency is for, and what a human being is. An AI model trained on human-generated text carries the values — including the pathologies — of the culture that produced it. A workflow that automates a customer interaction carries a design choice about whether that customer deserves a human being's attention. These are not engineering questions. They are moral ones.

The animating verse of this course is Genesis 2:15: "The Lord God took the man and put him in the Garden of Eden to work it and take care of it." To tend and to keep. Not to extract. Not to optimize without limit. Not to deploy at scale without accountability for what is broken. The imago Dei — the idea that human beings bear the image of a creating God — is not dissolved by the arrival of machines that can also create. It is clarified. If we are made to create, we are responsible for what we create. Including our AI systems.

This book will not offer that framing as optional. It is woven through Chapter 10's treatment of algorithmic accountability, Chapter 11's theology of delegation, and the ethics audit template you will complete before the semester ends. Students from all faith traditions are welcome to engage critically with this framing. Engagement — including disagreement — is the point.

How to Use This Book

The Labs: Every chapter includes a thirty-minute lab. Do them. Every one. The labs are where the concepts become capability. Reading about meta-prompting produces knowledge; writing a meta-prompt produces skill. These are not the same thing, and only one of them compounds.

The AI Studio Builds: Each week includes a hands-on build session in Google AI Studio or an equivalent environment. Come with a working product idea from Chapter 2 onward. The builds are cumulative — you are constructing a real system across the semester, not completing isolated exercises.

The Meta-Prompting Requirement: Every major deliverable in this course requires a submitted meta-prompt: a prompt you designed to generate or refine your output, with a written reflection on how it performed and how you iterated. This is not busywork. The ability to design effective prompts — to specify context, persona, constraints, and output format with precision — is a core professional skill in an AI-native organization. Appendix C provides templates. Use them as a starting point, not a shortcut.

The Discussion Boards: The discussion prompts in this course are not comprehension checks. They are structured disagreements. Come prepared to defend a position and change your mind when the argument warrants it. The best thinking in this course will happen in those exchanges.

A Note on Why This Was Written

This book was written because the textbooks that existed were not adequate for this moment. They were written for a world in which AI was a subject to study rather than an environment to operate in. Students deserve better than that — they deserve a course that treats them as future architects of organizations, not passive recipients of technology trends.

The goal is not to make you fluent in today's tools. Tools change. The goal is to make you structurally capable: someone who understands how AI systems create and destroy economic value, who can build and govern them responsibly, and who carries into their career a framework for evaluating whatever comes next. That framework — strategic, technical, and ethical — is what this book attempts to provide.

Do the labs. Build the system. Examine the values embedded in what you build.

Then go build something worth building.

Dr. Ernesto Lee Palm Beach Atlantic University 2025